

Lecture Scribe
Marginal PMF, Correlation, Covariance, Joint
Gaussian Random Variables
(Based strictly on Lecture-18 slides)

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1 1. Marginal PMF

1.1 Definition

For discrete random variables X and Y , the joint PMF is

$$p_{XY}(x, y) = P(X = x, Y = y)$$

The marginal PMF is obtained by summing the joint PMF over all possible values of the other variable.

Thus,

$$p_X(x) = \sum_y p_{XY}(x, y) \quad p_Y(y) = \sum_x p_{XY}(x, y)$$

Key Idea:

Marginal PMF of one variable = sum of joint PMF across all values of the other variable.

(Explained in slide definition)

1.2 Marginal PMF from Joint PMF Table

From the joint table:

Row sums $\rightarrow$ give $p_X(x)$

Column sums $\rightarrow$ give $p_Y(y)$

Mathematically,

$$p_X(x_i) = \sum_j p_{ij} \quad p_Y(y_j) = \sum_i p_{ij}$$

Also,

$$\sum_i p_X(x_i) = 1, \quad \sum_j p_Y(y_j) = 1$$

(as marginal PMFs must sum to 1).

1.3 Solved Example

Joint PMF:

$p_{XY}(x, y)$	$y = 0$	$y = 1$
$x = 0$	1/8	1/4
$x = 1$	3/8	1/4

Step-1: Find $p_X(x)$

$$p_X(0) = \frac{1}{8} + \frac{1}{4} = \frac{1}{8} + \frac{2}{8} = \frac{3}{8} \quad p_X(1) = \frac{3}{8} + \frac{1}{4} = \frac{3}{8} + \frac{2}{8} = \frac{5}{8}$$

Step-2: Find $p_Y(y)$

$$p_Y(0) = \frac{1}{8} + \frac{3}{8} = \frac{4}{8} = \frac{1}{2} \quad p_Y(1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Thus,

$$p_X(x) = \begin{cases} 3/8, & x = 0 \\ 5/8, & x = 1 \end{cases} \quad p_Y(y) = \begin{cases} 1/2, & y = 0 \\ 1/2, & y = 1 \end{cases}$$

(Example derivation shown in slides)

2. Correlation

2.1 Product Moment

A basic measure of joint behaviour is

$$R_{XY} = E[XY]$$

It represents the average value of the product XY .

Interpretation

Depends on scale of X and Y

Computed about origin (not about mean)

Mixes:

mean

spread

dependence

If

$$R_{XY} = 0$$

→ variables are orthogonal about origin

To remove mean effect → center variables → covariance.

3. Covariance

3.1 Definition

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Where

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

Interpretation

Covariance measures whether deviations from mean occur:

Same direction → Positive covariance

Opposite direction → Negative covariance

No linear relation $\rightarrow$ Zero covariance
 Magnitude depends on units $\rightarrow$ cannot measure strength.
 (Interpretation slide)

3.2 Algebraic Form (Step-by-Step Derivation)

Start with:
 $\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$
 Expand:
 $= E[XY - X\mu_Y - \mu_X Y + \mu_X \mu_Y]$
 Apply expectation:
 $= E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X \mu_Y$
 Since:
 $E[X] = \mu_X, \quad E[Y] = \mu_Y$
 Therefore:
 $\text{Cov}(X, Y) = E[XY] - \mu_X \mu_Y$
 (Full derivation from slide)

3.3 Properties

Symmetry

$$\text{Cov}(X, Y) = \text{Cov}(Y, X)$$

Variance special case

$$\text{Cov}(X, X) = \text{Var}(X)$$

Linearity

$$\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y)$$

Independence

$$X \perp Y \Rightarrow \text{Cov}(X, Y) = 0$$

But

Zero covariance $\neq$ independence (general case).

(Properties slide)

3.4 Solved Example-1

Given:

$$Y = -2X + 3$$

Find $\text{Cov}(X, Y)$

Using:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Step-1: Compute XY

$$XY = X(-2X + 3) = -2X^2 + 3X$$

Step-2: Compute $E[Y]$

$$E[Y] = E[-2X + 3] = -2E[X] + 3$$

Step-3: Substitute

$$\begin{aligned} \text{Cov}(X, Y) &= E[-2X^2 + 3X] - E[X](-2E[X] + 3) \\ &= -2E[X^2] + 3E[X] + 2(E[X])^2 - 3E[X] \end{aligned}$$

$$= -2E[X^2] + 2(E[X])^2$$

Since

$$Var(X) = E[X^2] - (E[X])^2$$

Thus,

$$Cov(X, Y) = -2Var(X)$$

(Example slide derivation)

3.5 Pearson Correlation Coefficient

$$\rho_{XY} = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}} = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$$

Interpretation

Measures strength + direction of linear relationship

Normalized covariance

Unit-free

Range:

$$-1 \leq \rho_{XY} \leq 1$$

$> 0 \rightarrow$ positive trend

$< 0 \rightarrow$ negative trend

$= 0 \rightarrow$ no linear correlation

(Slides explanation)

3.6 Solved Example-2 (Continuous Case)

Joint PDF:

$$f_{XY}(x, y) = \frac{xy}{96}, \quad 0 < x < 4, \quad 1 < y < 5$$

Step-1: Compute $E[X]$

$$E[X] = \int_0^4 xf_X(x)dx = \frac{1}{8} \int_0^4 x^2 dx$$

$$= \frac{1}{8} \left[\frac{x^3}{3} \right]_0^4 = \frac{1}{8} \cdot \frac{64}{3} = \frac{8}{3}$$

Step-2: Compute $E[Y]$

$$E[Y] = \int_1^5 yf_Y(y)dy = \frac{1}{12} \int_1^5 y^2 dy$$

$$= \frac{1}{12} \left[\frac{y^3}{3} \right]_1^5 = \frac{31}{9}$$

Step-3: Compute $E[XY]$

Since X, Y independent:

$$E[XY] = E[X]E[Y] = \frac{8}{3} \cdot \frac{31}{9} = \frac{248}{27}$$

Step-4: Linear Expectation

$$E[2X + 3Y] = 2E[X] + 3E[Y] = \frac{16}{3} + \frac{31}{3} = \frac{47}{3}$$

Step-5: Variance of X

$$E[X^2] = \frac{1}{8} \int_0^4 x^3 dx = 8$$

$$Var(X) = 8 - \left(\frac{8}{3}\right)^2 = \frac{8}{9}$$

Step-6: Variance of Y

$$E[Y^2] = 13$$

$$Var(Y) = 13 - \left(\frac{31}{9}\right)^2 = \frac{92}{81}$$

Step-7: Covariance

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = \frac{248}{27} - \frac{248}{27} = 0$$

Step-8: Correlation

$$\rho_{XY} = \frac{0}{\sqrt{\text{Var}(X)\text{Var}(Y)}} = 0$$

(All steps from solution slides)

4 4. Joint Gaussian Random Variables

4.1 Definition

A pair (X, Y) is jointly Gaussian if:

Joint density has bivariate Gaussian form

Marginals are Gaussian

Applications:

Signal processing

Communication systems

Noisy measurements modelling

(Definition slide)

4.2 Algebraic Form

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right)\right]\right)$$

Where:

$\mu_X, \mu_Y \rightarrow$ means

$\sigma_X, \sigma_Y \rightarrow$ standard deviations

$\rho \rightarrow$ correlation parameter

4.3 Interpretation Example

Example given:

X : CO concentration

Y : Global temperature anomaly

If:

$\rho > 0$

$\rightarrow$ High CO tends to occur with higher temperature.

(Interpretation slide)